



MIR Spectra and Analysis of Jovian Trojan Asteroids

Audrey C. Martin¹ and Joshua P. Emery² ¹ University of Central Florida, Department of Physics, 4111 Libra Drive, Physical Sciences Bldg. 430, Orlando, FL 32816, USA; Audrey.martin@ucf.edu² Northern Arizona University, Department of Astronomy and Planetary Science, 527 S. Beaver Steet, Flagstaff, AZ 86011, USA

Received 2023 June 18; revised 2023 July 31; accepted 2023 August 1; published 2023 August 29

Abstract

Jovian Trojan asteroids make up a large group of primitive bodies that populate Jupiter’s L4 and L5 Lagrange points. The Trojans’ dynamics and composition carry insight into the formation mechanisms that shaped our solar system. Mid-infrared (MIR; 5–35 μm) spectra of Trojans exhibit puzzling silicate emission features, like spectra of optically thin comet comae, which may be interpreted as “fluffy” regoliths. By understanding the physical properties of the regolith (e.g., particle size and regolith porosity), more accurate compositional interpretations can be made. Here we show 14 Spitzer Space Telescope MIR spectra of Trojans and their compositional makeup. Through parameterization of spectral features and comparison to laboratory and remote-sensing spectra, the results show that the Trojans have highly porous regoliths of fine-particulate crystalline (forsterite, enstatite, and possibly a small amount of fayalite and diopside) and amorphous silicates. These results are consistent with a primordial Kuiper Belt origin.

Unified Astronomy Thesaurus concepts: Asteroids (72); Small Solar System bodies (1469); Trojan asteroids (1715); Asteroid surfaces (2209); Spectral energy distribution (2129); Infrared spectroscopy (2285); Spectroscopy (1558)

1. Introduction

Jovian Trojan asteroids (hereafter Trojans) populate Jupiter’s L4 and L5 Lagrange points, which are dynamically stable over the lifetime of the solar system (Levison et al. 1997). As a large population of primitive asteroids that orbit in a unique position, Trojans tie together solar system formation theories and dynamical models. However, compositional studies that incorporate the Trojans’ potential dynamical history remain ambiguous.

Trojans are thought to have formed in one of two locations: (1) their current position (~ 5.2 au; Marzari & Scholl 1998; Fleming & Hamilton 2000) or (2) the primordial Kuiper Belt (~ 15 –30 au; Morbidelli et al. 2005). The in situ formation hypothesis describes a situation where Trojans were captured into Jupiter’s Lagrange points as Jupiter grew in the early stages of planet formation. As Jupiter grows, the regions of stability around the Lagrange points grow, and asteroids in the area become “locked” in L4 and L5 (Marzari & Scholl 1998; Fleming & Hamilton 2000). The alternative Kuiper Belt origin is motivated by the Nice model, which describes giant planet migration (Gomes et al. 2005; Morbidelli et al. 2005; Tsiganis et al. 2005). In the Nice model, the giant planets originate with heliocentric distances of < 20 au, and the primordial Kuiper Belt orbited in Neptune’s and Uranus’s current region. Eventually, the giant planet orbits were destabilized, causing Neptune and Uranus to migrate into the primordial Kuiper Belt, which scattered Kuiper Belt objects (KBOs) all over the solar system. Simulations show that Jupiter’s Lagrange points became unstable, lost all preexisting Trojans, remained unstable during giant planet migration, and became stable once migration ended (Morbidelli et al. 2005; Nesvorný et al. 2013). Thus, any scattered KBOs that were in Jupiter’s stable

Lagrange points directly after giant planet migration are still there (Morbidelli et al. 2005; Levison et al. 2008).

Primitive asteroid compositions can be indicative of formation regions, as the inner solar system tended to be rich in Mg-rich crystalline silicates, while the outer solar system tended to be rich in Fe-rich silicates (Gail 2004; Henning 2010). Furthermore, had the Trojans accreted in the inner solar system (inward of ~ 13 au), it is likely that they would show evidence of aqueous alteration. Many middle and outer main-belt asteroids (MBAs) are composed of hydrated silicates, carbon-bearing compounds (Gaffey et al. 1993; Hiroi et al. 1996; Rivkin et al. 2003, 2006; Clark et al. 2009; Takir & Emery 2012; DeMeo & Carry 2013), and some anhydrous silicates (Vernazza et al. 2017). Ice-rich asteroids that accreted quickly (i.e., early in the protoplanetary disk) would have incorporated enough ^{26}Al to melt ice and drive aqueous alteration (Grimm & McSween 1989). Asteroids that accreted later or in water-free zones would remain largely anhydrous. Spectroscopy can offer compositional clues that get to the heart of the Trojan origins.

Trojan surfaces have low albedos and red-sloped spectra in the visible and near-infrared (VNIR; 0.8–4.0 μm) with no diagnostic absorption features (Jewitt & Luu 1990; Cruikshank et al. 2001; Emery & Brown 2003; Fernández et al. 2003; Emery et al. 2011; Yang & Jewitt 2011; Romanishin & Tegler 2018). Trojans fall into two distinct groups based on spectral slope: red-sloped and less red-sloped (Emery et al. 2011). The spectral slopes of both groups can be explained by anhydrous silicates (Emery & Brown 2004) rather than the organic materials that were originally hypothesized (Gradie & Veverka 1980).

Analysis of mid-infrared (MIR; 5–30 μm) spectra shows that the surfaces of three Trojans (624 Hektor, 1172 Aneas, and 911 Agamemnon, all of which are in the red VNIR spectral group) have emissivity features like comet comae made of fine-particulate silicates (Emery et al. 2006). Typically, the MIR spectra of regoliths exhibit emissivity minima near 10 and



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](https://creativecommons.org/licenses/by/4.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Table 1
Observing Parameters for Trojan Targets

Object	Date (UT)	Time (UT)	Exposure Time (s)				t_{AOR} (hr)	r (au)	Δ (au)	α (deg)
			SL2	SL1	LL2	LL1				
588 Achilles	13.06.07	9:23:48	1920	1200	56	24	1.198	5.6650	5.5463	10.30
617 Patroclus	24.06.07	20:58:02	1440	840	36	24	0.906	5.8916	5.3703	08.88
659 Nestor	12.06.07	23:04:33	1920	480	56	24	0.947	4.6319	4.4123	12.56
1437 Diomedes	16.11.06	09:00:37	960	56	24	24	0.465	5.2488	4.8756	10.76
1867 Deiphobus	11.06.07	19:14:53	960	240	24	24	0.524	5.1314	4.5519	09.85
2797 Teucer	12.06.07	22:11:55	1440	360	36	24	0.737	5.5349	5.3729	10.52
3548 Eurybates	16.11.06	13:16:50		2400	240	120	1.029	5.3432	5.2715	10.99
4035 Thestor	13.06.07	08:12:35		2400	300	120	1.051	5.3779	5.3381	10.84
4060 Deipylos	13.06.07	07:12:16		1920	300	120	0.883	6.0147	5.8900	09.69
4709 Ennomos	28.06.06	23:00:17	960	960	84	36	0.791	5.2915	5.0324	10.93
11395 Iphinous	19.06.07	23:51:46		1920	240	120	0.862	4.8617	4.8003	12.03

Note. Values for the beginning of each observation are listed for time-specific parameters. The t_{AOR} is the total length of time, including overhead. The parameters r , Δ , and α are the heliocentric distance, SST object distance, and SST-centered phase angle, respectively.

20 μm due to surface scattering in the Si–O fundamental vibrational bands (e.g., Christensen et al. 2000; Donaldson Hanna et al. 2017). However, Trojan emissivity spectra have peaks at these wavelengths, indicating that the regolith configuration supports dominant primary emission and minimal contribution from surface scattering (e.g., Hunt & Logan 1972). Researchers hypothesize that the anomalous MIR spectra indicate that Trojans have a “fluffy” surface, and laboratory investigations back up this theory (e.g., Vernazza et al. 2012; Martin et al. 2022, 2023).

The MIR spectral region is particularly sensitive to the mineralogy and composition of silicates. For example, the precise band position of silicates like olivine and pyroxene (abundant minerals in the solar system) increases linearly as Mg is replaced with Fe in the crystal lattice (e.g., Koike et al. 2003; Hofmeister & Pitman 2007; Hamilton 2010; Lane et al. 2011). Thus, MIR spectra are useful for determining the mineralogy and Mg# (i.e., $\text{Mg}/[\text{Fe}+\text{Mg}]$) of anhydrous silicates. Furthermore, determining these compositional aspects of Trojan surfaces is useful to constrain the formation region. In this paper, we present new spectra of 11 Trojans, determine the composition and regolith properties of all 14 Trojans that have been observed in the MIR, and place our results in the context of Trojan formation theories.

2. Target, Observations, and Data Reduction

Measurements of the 11 new Trojan spectra were made between 2006 July and 2007 July using the Spitzer Space Telescope (SST; Werner et al. 2004), program number 30228. Trojans (617) Patroclus, (1867) Deiphobus, and (4709) Ennomos are found in the L5 swarm, while the rest are in L4. The red Trojans are (588) Achilles, (1867) Deiphobus, (2797) Teucer, and (4035) Thestor, and the remainder are less red. All data were collected in Infrared Spectrograph (IRS; Houck et al. 2004) stare mode while the instrument was actively cooled with liquid He. In low spectral resolution mode ($R = \lambda/\Delta\lambda \approx 64\text{--}128$), the IRS instrument measured the 5.2–38 μm spectral range along four slit segments: short wavelength, low resolution, second order (SL2; 5.2–8.5 μm); short wavelength, low resolution, first order (SL1; 7.4–14.2 μm); long wavelength, low resolution, second order (LL2; 14.0–21.5 μm); and long wavelength, low resolution, first order (LL1; 19.5–38.0 μm). Each object was observed in two dither positions along the slit.

See Table 1 for a detailed summary of the objects and observing parameters. The asteroids (3548) Eurybates, (4035) Thestor, (11395) Iphinous, and (4060) Deipylos were observed using SL1, LL2, and LL1 (i.e., 7.4–38 μm). The remaining targets were observed with all four segments (Table 1).

Our data reduction methods closely follow those described in Emery et al. (2006), beginning with basic calibrated data (BCD) files produced by the Spitzer Science Center (SSC) on the Spitzer heritage archive³ (pipeline version S18.18). The BCDs are dark current-subtracted and stray light- and flat field-corrected, bad pixels have been flagged, and the IRS 3D data file has been collapsed into a 128×128 2D flux image (IRS Instrument Handbook). To reduce residual background emission, we take the difference between two dither positions. Targets are identified, and their 1D spectra are extracted within the slit by applying a spatial Gaussian, removing outliers, fitting the result to a second-order polynomial, and finally summing the flux in apertures at each wavelength with widths appropriate for the point-spread function (PSF). The PSF widths are wavelength-dependent because Spitzer is diffraction-limited. So, we use the SSC extraction width (7''/2, 14''/4, 21''/68, and 36''/58 at 6, 12, 16, and 27 μm , respectively) and then linearly modify the width with wavelength. The four slit orders are scaled by fitting a polynomial to the ends of adjacent orders and manually adjusting for continuity. Finally, the whole spectrum is rescaled to the mean of each order’s respective scale factor.

3. Thermal Modeling

Thermal modeling begins with the spectral energy distribution (SED). The SED is affected by the size, composition, and temperature distribution of an object. To model the SED, we use the Near-Earth Asteroid Thermal Model (NEATM; Harris 1998), a modified standard thermal model, as is common for many non-near-Earth asteroid studies (e.g., Emery et al. 2006; Lim et al. 2011; Licandro et al. 2012). With the NEATM, we derived the subsolar temperature (T_{SS}) and estimated the radius (R), visible geometric albedo (p_v), and beaming parameter (η) for each object (Table 2).

NEATM assumes a spherical shape and radiative equilibrium at the surface of the asteroid (i.e., heat conduction into

³ <https://sha.ipac.caltech.edu/applications/Spitzer/SHA/>

Table 2
NEATM Best-fit Parameter Results

Object	H_v	R (km)	p_v	η	T_{ss} (K)	Grav et al. (2011)		
						R (km)	p_v	η
Achilles	8.67	$73.005^{+2.990}_{-2.557}$	$0.028^{+0.004}_{-0.004}$	$0.809^{+0.022}_{-0.002}$	$178.7^{+1.2}_{-1.3}$	80.30 ± 5.95	0.023 ± 0.006	1.022 ± 0.092
Patroclus	8.19	$76.679^{+2.893}_{-3.164}$	$0.040^{+0.004}_{-0.007}$	$0.826^{+0.024}_{-0.036}$	$174.1^{+1.3}_{-1.9}$	92.55 ± 6.55	0.027 ± 0.006	0.897 ± 0.076
Nestor	8.99	$60.455^{+3.253}_{-3.615}$	$0.031^{+0.003}_{-0.007}$	$0.958^{+0.066}_{-0.078}$	$189.4^{+3.4}_{-3.9}$	61.20 ± 5.35	0.030 ± 0.007	0.931 ± 0.101
Diomedes	8.3	$80.244^{+2.929}_{-2.811}$	$0.033^{+0.004}_{-0.005}$	$0.725^{+0.012}_{-0.001}$	$190.7^{+0.8}_{-0.2}$	64.04 ± 4.90	0.052 ± 0.012	1.003 ± 0.096
Deiphobus	8.3	$69.112^{+2.810}_{-3.065}$	$0.044^{+0.004}_{-0.008}$	$0.822^{+0.024}_{-0.034}$	$186.7^{+1.4}_{-1.8}$	72.64 ± 3.88	0.040 ± 0.008	1.057 ± 0.067
Teucer	8.8	$56.573^{+2.146}_{-1.984}$	$0.042^{+0.005}_{-0.006}$	$0.744^{+0.019}_{-0.003}$	$184.4^{+1.1}_{-0.2}$			
Eurybates	9.7	$35.119^{+1.323}_{-1.475}$	$0.047^{+0.005}_{-0.008}$	$0.899^{+0.016}_{-0.032}$	$178.9^{+0.9}_{-1.5}$	29.68 ± 0.74	0.060 ± 0.016	0.780 ± 0.027
Thestor	9.6	$36.238^{+1.373}_{-1.263}$	$0.049^{+0.006}_{-0.007}$	$0.813^{+0.018}_{-0.008}$	$182.8^{+1.0}_{-0.4}$	30.90 ± 0.53	0.067 ± 0.012	0.737 ± 0.017
Deipylus	9.4	$45.132^{+3.464}_{-4.070}$	$0.038^{+0.004}_{-0.011}$	$0.965^{+0.090}_{-0.113}$	$165.8^{+4.0}_{-5.0}$			
Ennomos	8.6	$48.788^{+1.955}_{-1.765}$	$0.067^{+0.009}_{-0.010}$	$1.100^{+0.034}_{-0.019}$	$170.6^{+1.3}_{-0.6}$	41.90 ± 3.68	0.091 ± 0.024	0.897 ± 0.101
Iphinous	9.8	$35.858^{+3.223}_{-3.587}$	$0.041^{+0.006}_{-0.014}$	$0.875^{+0.113}_{-0.130}$	$188.9^{+6.4}_{-7.3}$	36.37 ± 6.65	0.040 ± 0.013	0.703 ± 0.166

Note. The table includes a comparison to physical properties derived from WISE thermal observations from Grav et al. (2011).

the subsurface is ignored, or, equivalently, the thermal inertia is assumed to be zero). These assumptions lead to a simple expression for the temperature at the subsolar point of the asteroid,

$$T_{ss} = \left[\frac{S_0(1 - A_B)}{r_{AU}^2 \sigma \varepsilon \eta} \right]^{1/4}, \quad (1)$$

where S_0 is the solar flux at 1 au (1374 W m^{-2}), A_B is the bolometric Bond albedo, r_{AU} is the distance between the Sun and the object in astronomical units, σ is the Stefan–Boltzmann constant ($5.67 \times 10^{-8} \text{ W m}^{-2}$), and ε is the bolometric emissivity (assumed here to be 0.9 for all objects). The A_B , r_{AU} , and visible absolute magnitude (H_v), which is used to compute p_v from the derived diameter, were obtained from JPL Horizons.⁴ Temperature varies across the asteroid’s surface as $\cos^{1/4}\theta_i$, where θ_i is the incidence angle. The flux at a particular wavelength is calculated by integrating the Planck function over the object’s observed hemisphere (with the temperature assumed to be zero on the nightside). In order to calculate the thermal continua for the objects, physical properties (R and η) are allowed to vary such that χ^2 is minimized, and the best-fit parameters are chosen. Figure 1 shows the measured spectral flux and the best-fit thermal continuum for each object.

4. Emissivity Spectra

4.1. Spectral Description

To compute the emissivity spectra, we divided the SED by the best-fit NEATM result and multiplied by ε . The 11 spectra bear strong resemblances to each other and the spectra of Hektor, Aneas, and Agamemnon (Figure 2). At short wavelengths, the spectra, excluding objects with no SL2 data and Ennomos, exhibit a sharp emissivity low around $7 \mu\text{m}$ followed by a peak at $\sim 7.5 \mu\text{m}$ and another dip near $8 \mu\text{m}$. The second dip is much shallower than the $7 \mu\text{m}$ dip and varies in depth from Diomedes to Deiphobus. At longer wavelengths, the spectra all show a plateau from ~ 9 to $\sim 11 \mu\text{m}$, which we refer to as the “ $10 \mu\text{m}$ plateau,” with a variety of shapes. Near $13 \mu\text{m}$, each spectrum exhibits a small emissivity peak in the center of an overall emissivity dip. The dip is followed by two

broad peaks centered at ~ 18 and $25 \mu\text{m}$, with a dip near $20 \mu\text{m}$ in between. The signal-to-noise ratio at wavelengths longer than $\sim 32 \mu\text{m}$ is too low to identify any features.

4.2. Parameterization

Silicate spectra in the MIR are highly dependent on factors such as composition (e.g., Mg#), particle size, regolith porosity, albedo, temperature, temperature gradient, and pressure (e.g., Vincent & Hunt 1968; Salisbury & Wald 1992; Mustard & Hays 1997; Hamilton 2010; Martin et al. 2022, 2023). These dependencies are particularly sensitive in the centers of Si–O stretching and bending vibrational modes in the ~ 10 and $18 \mu\text{m}$ regions, respectively. As such, we focus our spectral analysis on the $10 \mu\text{m}$ plateau and other silicate features. A visualization of the $10 \mu\text{m}$ plateau parameters is shown in Figure 3, using Deiphobus as an example. Note that we denote emissivity dips with “D” and peaks with “P.”

To calculate the position of each feature, $10 \mu\text{m}$ plateau parameters, and the error associated with each value, we create 10,000 synthetic versions of each Trojan spectrum. Each synthetic spectrum is created by selecting a randomly assigned emissivity value within the bounds of error at each wavelength. Then a fourth-order smoothing spline polynomial is fit to each synthetic spectrum. For each feature, we isolate a small segment of the polynomial in the approximate region of the feature and identify the local maximum or minimum. The mean position and standard deviation of each feature in a set of 10,000 polynomials are used as the final feature position and error, respectively. For example, to locate D_{12} , we isolate a segment of the spectrum from the approximate position of P_{10} and the small peak between D_{12} and D_{13} . Position results are listed in Table 3. In addition to calculating common broad features in each Trojan spectrum, we found small-scale features that superpose the $10 \mu\text{m}$ plateau that are useful for mineralogical analysis utilizing the same methods, as well as searching for features by eye (see Table 5).

We additionally calculated the spectral contrast, width, slope, skew, and peak position in continuum-removed space of the $10 \mu\text{m}$ plateau, as its shape can be indicative of not only mineralogy but also particle size and regolith porosity (e.g., Martin et al. 2022, 2023). The error was estimated using the same synthetic spectrum method as before. To calculate the parameters, we used D_8 and D_{12} to bound the $10 \mu\text{m}$ plateau

⁴ <https://ssd.jpl.nasa.gov/horizons/app.html#/>

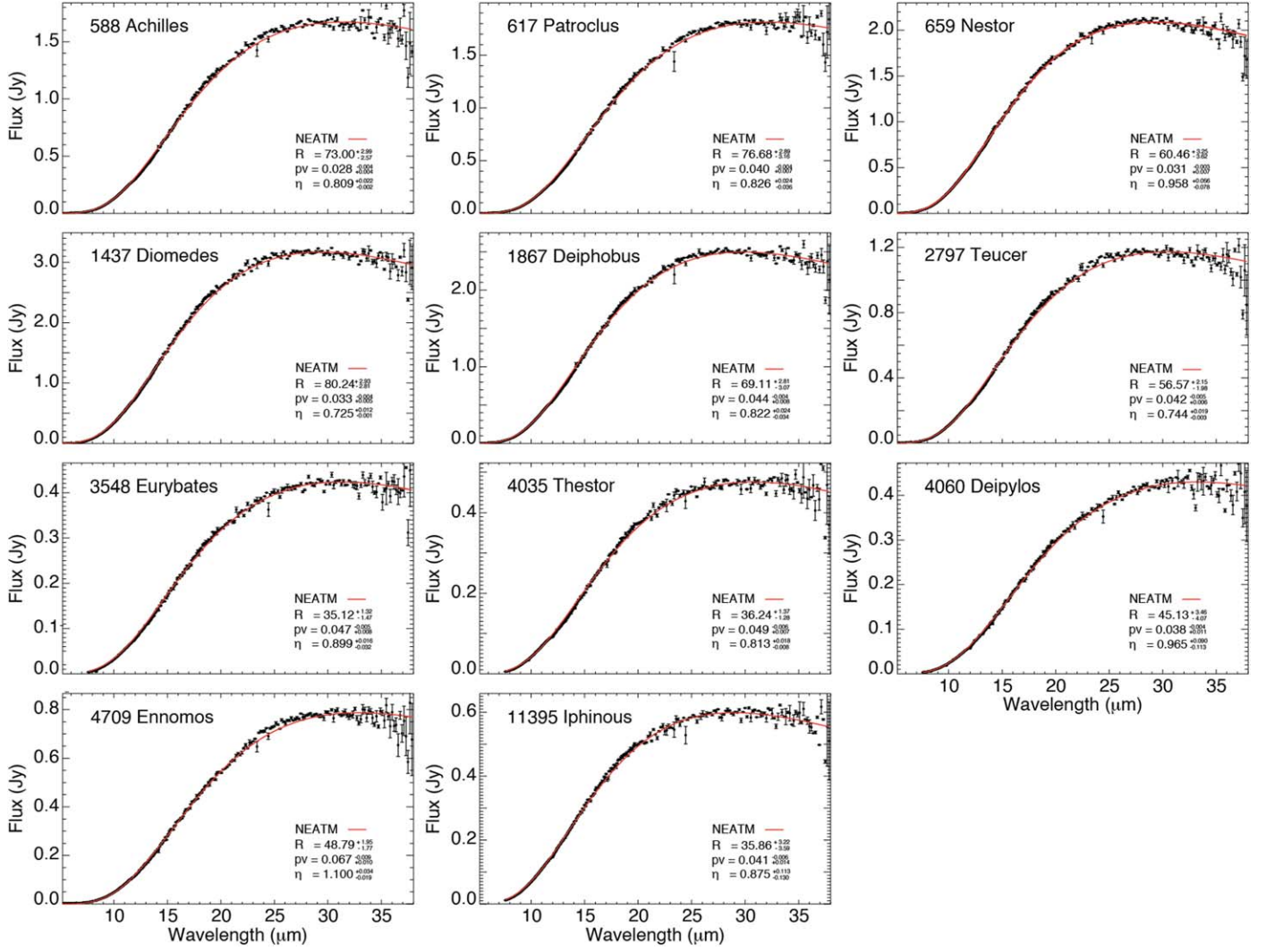


Figure 1. Measured SEDs (black points) of the Trojans with the calculated best-fit NEATM (red lines) for each object. The radius, p_v , and η for each Trojan is listed in its respective panel and Table 2.

and calculated the slope between the two points as the continuum and the distance between them as the width. We divide the emissivity values of the $10\ \mu\text{m}$ plateau by a linear continuum fit between the two points. We calculate the spectral contrast by taking the difference between the peak of the $10\ \mu\text{m}$ plateau and the continuum. The peak wavelength will not necessarily be the same in continuum-removed space. Additionally, the $10\ \mu\text{m}$ plateau tends to lean toward longer wavelengths in both continuum-removed and non-continuum-removed space, which we define as the “skew” of the $10\ \mu\text{m}$ plateau. We measure the skew by calculating the difference between P_{10} and the midway point of the $10\ \mu\text{m}$ plateau (i.e., midway between D_8 and D_{12}) and converting the difference into percent of the width,

$$\left([P_{10} - (D_8 + \frac{\text{width}}{2})] / \text{width} \right) * 100. \quad (2)$$

A skew of 0% means that P_{10} falls directly over the midway point of the $10\ \mu\text{m}$ plateau, a positive skew indicates that the plateau leans toward longer wavelengths, and a negative skew means that the plateau leans toward shorter wavelengths. Interestingly, all of the Trojans skew toward longer wavelengths, except for Aneas, whose error overlaps with zero.

Trojans that fall into the “red” category have higher spectral contrasts compared to the “less-red” Trojans (as defined in Emery et al. 2011). Interestingly, the spectral contrasts are not in two separate groups; they fall along a continuum (Figure 4). Achilles has the lowest spectral contrast of the red Trojans with $7.04\% \pm 0.39\%$, while Iphinous has the highest spectral contrasts among the less-red Trojans with $6.54\% \pm 0.47\%$. The difference between these two ($\sim 0.5\%$) is less than the range of values within the red ($\sim 3.6\%$) and less-red ($\sim 2.3\%$) groups, respectively (see Table 4). We did not find any other correlations between the parameters we measured or values we calculated (object radius, temperature, and albedo). Additional relationships between parameters (e.g., skew versus width) were explored, but we did not find any significant relationships.

5. Comparisons to Potential Analogs

5.1. Regolith Porosity

As regolith porosity strongly affects the shape and sometimes position of important spectral bands, it is necessary to establish the regolith porosity of each Trojan surface before comparing Trojan spectra to potential end-members from spectral libraries. Martin et al. (2022, 2023) concluded that

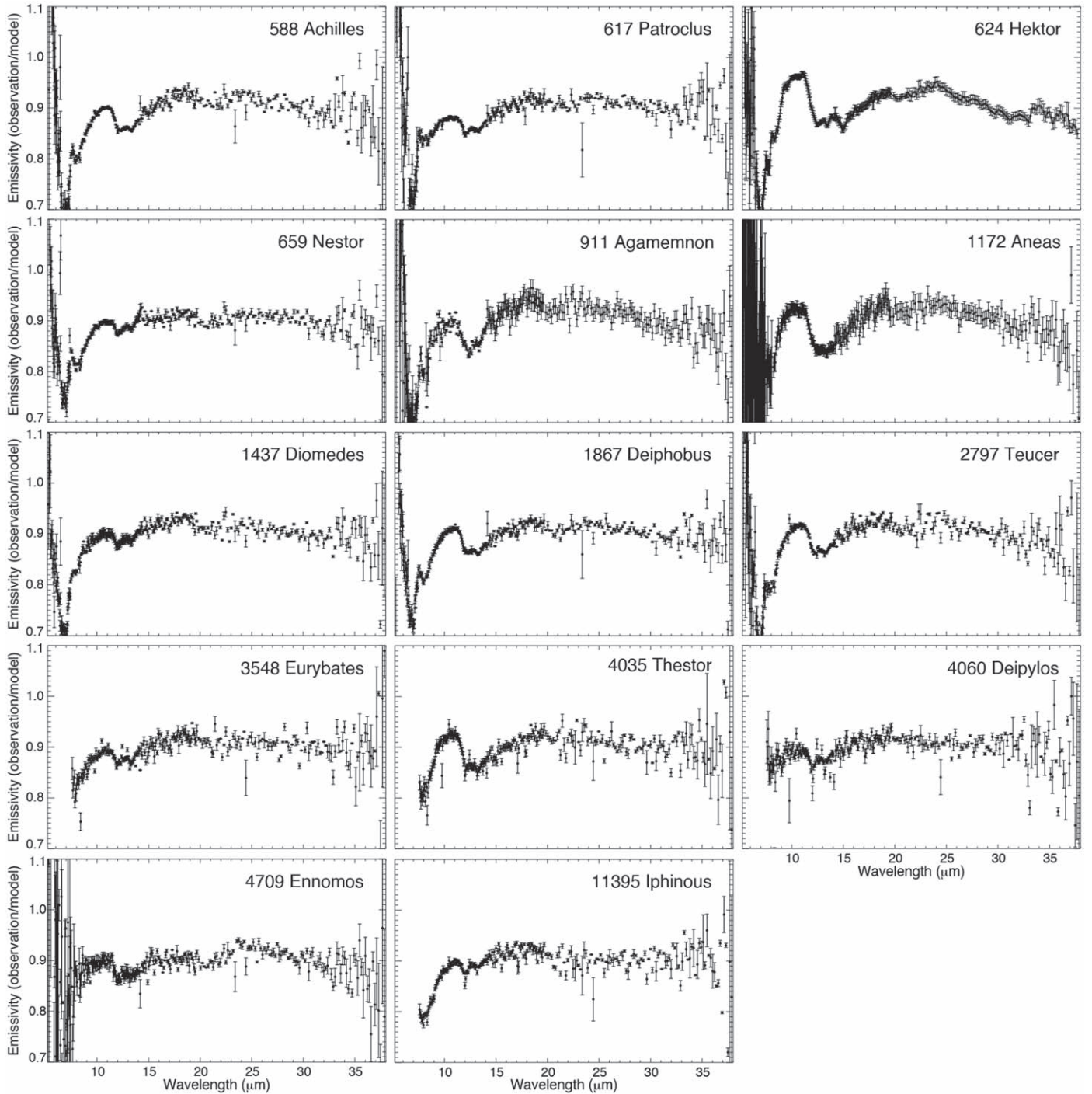


Figure 2. The SST/IRS emissivity spectra of Achilles, Patroclus, Nestor, Diomedes, Deiphobus, Teucer, Eurybates, Thestor, Deipylos, Ennomos, and Iphinous. The spectra of Hektor, Aneas, and Agamemnon from Emery et al. (2006) have been rereduced using version S18.18 of the Spitzer calibration pipeline.

Hektor’s regolith porosity is at least 80% (i.e., at least 80% of Hektor’s regolith is void space); here we use the same lines of evidence to conclude that all of our Trojans have high regolith porosities. There are two general lines of evidence for high regolith porosity: the absence of Christiansen features (CFs) and the position of crystalline peaks. A CF is a peak in emissivity where a strong band transitions from volume scattering along the electromagnetic spectrum (around $8.9\ \mu\text{m}$ in forsterite spectra; Christiansen et al. 2000).

Trojan spectra are consistent with high regolith porosity (e.g., $\gtrsim 80\%$) because they lack CFs and exhibit a forsterite

band at $\sim 11.1\ \mu\text{m}$. Silicate spectra primarily in the surface scattering regime have CFs around $8.9\ \mu\text{m}$. Aside from Eurybates, we have not identified an $8.9\ \mu\text{m}$ feature in any Trojan spectra. However, it is important to note that the Eurybates spectrum is quite noisy in the CF region. For small particles ($< 20\ \mu\text{m}$), the CF disappears around 80% total regolith porosity (void space plus any transparent material) and would be consistent with any regolith porosity higher than 80% (Martin et al. 2022, 2023). For this same particle size, an $11\ \mu\text{m}$ olivine feature is at $11.04\ \mu\text{m}$ for regolith porosities of $\sim 80\%$ and shifts to $11.14\ \mu\text{m}$ when the regolith porosity

Table 3
Position of Each Feature in Trojan Spectra Given in Micrometers

Object	D ₇	D ₈	D ₁₂	D ₁₃	P ₁₀	P ₁₃	P ₁₈	P ₂₅
Achilles	6.98 ± 0.02	7.99 ± 0.01	12.19 ± 0.01	13.35 ± 0.05	11.05 ± 0.18	12.87 ± 0.02	18.20 ± 0.45	24.57 ± 0.33
Patroclus	6.98 ± 0.01	8.03 ± 0.05	12.07 ± 0.03	13.21 ± 0.14	10.59 ± 0.07	12.76 ± 0.11	18.56 ± 0.34	24.67 ± 0.25
Hektor	6.97 ± 0.01	8.50 ± 0.04	12.50 ± 0.02	14.89 ± 0.42	11.07 ± 0.02	13.03 ± 0.04	18.32 ± 0.16	23.94 ± 0.19
Nestor	6.98 ± 0.01	7.99 ± 0.01	12.06 ± 0.04	13.33 ± 0.07	10.68 ± 0.15	12.87 ± 0.05	18.03 ± 0.56	26.38 ± 2.30
Agamemnon	6.98 ± 0.01	8.22 ± 0.13	12.42 ± 0.04	13.31 ± 0.67	10.95 ± 0.47	12.91 ± 0.07	18.22 ± 0.86	22.86 ± 0.51
Aneas	N/A	8.00 ± 0.16	12.56 ± 0.01	13.57 ± 0.44	10.27 ± 0.28	12.99 ± 0.01	18.95 ± 0.42	24.09 ± 0.10
Diomedes	6.98 ± 0.02	7.86 ± 0.12	12.00 ± 0.02	13.31 ± 0.07	10.72 ± 0.29	12.86 ± 0.13	18.63 ± 0.20	24.74 ± 0.36
Deiphobus	6.98 ± 0.01	7.99 ± 0.01	12.23 ± 0.19	13.37 ± 0.27	11.08 ± 0.12	12.88 ± 0.23	18.41 ± 0.07	24.57 ± 0.33
Teucer	6.98 ± 0.01	7.80 ± 0.01	12.22 ± 0.04	13.22 ± 0.07	10.71 ± 0.18	12.77 ± 0.06	17.90 ± 0.46	23.93 ± 0.34
Eurybates	N/A	7.85 ± 0.10	12.00 ± 0.02	13.36 ± 0.03	10.51 ± 0.01	12.74 ± 0.02	18.48 ± 0.28	N/A
Thestor	N/A	7.85 ± 0.13	12.12 ± 0.01	13.20 ± 0.05	10.55 ± 0.07	12.59 ± 0.07	18.82 ± 0.41	N/A
Deipylus	N/A	8.03 ± 0.43	12.01 ± 0.02	13.20 ± 0.07	10.52 ± 0.47	12.49 ± 0.03	19.70 ± 0.22	N/A
Ennomos	6.98 ± 0.01	7.90 ± 0.21	11.95 ± 0.12	13.14 ± 0.19	10.75 ± 0.19	12.59 ± 0.17	19.51 ± 1.61	25.04 ± 0.85
Iphinous	N/A	7.88 ± 0.11	12.06 ± 0.02	13.16 ± 0.09	11.10 ± 0.06	12.56 ± 0.03	18.39 ± 0.09	N/A

Note. Here D_x refers to a dip near $x \mu\text{m}$, and P_y refers to a peak near $y \mu\text{m}$.

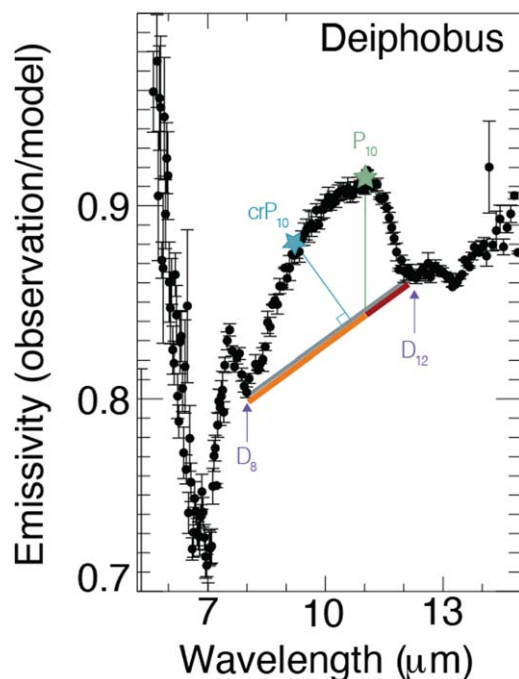


Figure 3. The MIR spectrum of *Deiphobus* with parameter annotation. The D_8 and D_{12} (purple) are labeled and marked with arrows. The continuum (gray) slopes upward from D_8 to D_{12} , which designate the start and end of the $10\text{ }\mu\text{m}$ plateau, respectively. The width is calculated from D_8 to D_{12} . The P_{10} (green) is the peak location given in Table 3, while $\text{cr}P_{10}$ (blue) is the $10\text{ }\mu\text{m}$ plateau peak in continuum-removed space listed in Table 4. The spectral contrast is calculated in continuum-removed space using $\text{cr}P_{10}$. Skew is the ratio between the right (orange) and left (red) sides of P_{10} as calculated in Equation (2).

increases to $\sim 95\%$. We identify an $11.1\,\mu\text{m}$ feature in the spectra of all Trojans except Eurybates and Nestor, which corresponds to regolith porosities of $\sim 90\%$, on average (Martin et al. 2022).

Since Eurybates may have a CF, the regolith porosity could be less than $\sim 80\%$. The CFs shift their position with changes in regolith porosity (Martin et al. 2022). We can estimate Eurybates’ regolith porosity using just the CF position. If we assume that the particle size on Eurybates is the same as on other Trojans, a CF near $8.9\,\mu\text{m}$ is about $\sim 80\%$ regolith porosity. Of note, the CF disappears at greater regolith

porosities in spectra of larger particles ($>20\ \mu\text{m}$), so 80% is a lower estimate. As such, all of the Trojan spectra show evidence of having highly fluffy surfaces of fine-particulate anhydrous silicates, $\sim 80\%$ on Eurybates and $\sim 90\%$ on the rest.

5.2. Individual Peak Fitting

Here we identify individual peaks in the 10 μm region for each Trojan to appropriately attribute potential end-members to specific features (Figure 5). Features were first identified using the methods outlined in Section 4.2 and, in some cases, by eye. Below, we discuss the possibility of spectral evidence of common astronomical minerals (olivine, pyroxene, phyllosilicates, and amorphous silicates) in Trojan spectra (see Table 6 for a summary). As Trojan spectra are consistent with high regolith porosity (volume scattering-dominated spectra), we use spectra from the appropriate spectral libraries for comparison (Koike & Shibai 1990; Koike et al. 2000, 2003; Chihara et al. 2002; Martin et al. 2022, 2023). In addition to the spectral comparisons detailed below, we performed linear deconvolutions and meteorite spectral comparisons and searched for the source of the P₁₃ feature to no avail (a detailed analysis can be found in Martin 2022, Sections 4.5.1 and 4.5.3).

5.2.1. Olivine

Olivine exists in a solid solution between forsterite (Mg-rich) and fayalite (Fe-rich). The two primary resonance bands in forsterite (~ 10.0 and $11.2 \mu\text{m}$) shift to longer wavelengths as Fe replaces Mg in the olivine mineral lattice (10.5 and $11.4 \mu\text{m}$; Figure 2 in Koike et al. 2003). Every Trojan spectrum except Nestor exhibits an $\sim 11.1 \mu\text{m}$ peak, which we interpret as evidence of forsterite (Table 5). Additionally, the presence of an $\sim 10.0 \mu\text{m}$ feature in the Hektor, Patroclus, Aeneas, Agamemnon, Eurybates, Thestor, and Deipylus spectra is further evidence of forsterite (Figure 5, Table 5).

All but two Trojan spectra show potential evidence for fayalite. We say “potential” because most fayalitic features are slightly better fits to enstatite (i.e., the wavelength and relative height of a given feature is better explained with a different mineral). Nevertheless, all but Deiphobus and Teucer have a peak near 10.5 μm , the primary resonance band for fayalite. Achilles, Aeneas, Agamemnon, Eurypates, Thestor, Deipylos,

Table 4
Calculated Parameter Values for the 10 μm Plateau

Object	crP ₁₀ (μm)	Spectral Contrast (%)	Width (μm)	Slope (μm^{-1})	Skew (%)
Achilles	10.43 $\pm$ 0.05	7.04 $\pm$ 0.39	4.19 $\pm$ 0.02	0.014 $\pm$ 0.001	22.88 $\pm$ 4.37
Patroclus	10.45 $\pm$ 0.11	4.27 $\pm$ 0.53	4.04 $\pm$ 0.05	0.004 $\pm$ 0.001	13.45 $\pm$ 2.16
Hektor	10.13 $\pm$ 0.09	11.77 $\pm$ 0.81	4.00 $\pm$ 0.04	0.005 $\pm$ 0.001	14.25 $\pm$ 0.23
Nestor	9.93 $\pm$ 0.17	5.14 $\pm$ 0.83	4.07 $\pm$ 0.04	0.016 $\pm$ 0.001	16.14 $\pm$ 3.68
Agamemnon	9.23 $\pm$ 0.23	11.75 $\pm$ 1.50	4.20 $\pm$ 0.11	0.012 $\pm$ 0.001	15.00 $\pm$ 1.52
Aneas	10.13 $\pm$ 0.05	10.36 $\pm$ 2.24	4.56 $\pm$ 0.16	0.009 $\pm$ 0.001	-0.22 $\pm$ 1.23
Diomedes	9.61 $\pm$ 0.70	4.54 $\pm$ 1.18	4.14 $\pm$ 0.13	0.011 $\pm$ 0.002	19.11 $\pm$ 7.74
Deiphobus	10.33 $\pm$ 0.19	7.62 $\pm$ 0.76	4.24 $\pm$ 0.19	0.013 $\pm$ 0.001	22.78 $\pm$ 3.77
Teucer	9.99 $\pm$ 0.22	8.69 $\pm$ 0.82	4.41 $\pm$ 0.04	0.016 $\pm$ 0.001	15.88 $\pm$ 4.18
Eurybates	10.50 $\pm$ 0.13	5.21 $\pm$ 0.82	4.14 $\pm$ 0.10	0.015 $\pm$ 0.004	14.10 $\pm$ 2.70
Thestor	10.48 $\pm$ 0.15	10.64 $\pm$ 1.57	4.27 $\pm$ 0.13	0.013 $\pm$ 0.003	13.24 $\pm$ 3.85
Deipylos	11.44 $\pm$ 0.11	5.63 $\pm$ 0.89	3.98 $\pm$ 0.44	0.002 $\pm$ 0.008	12.77 $\pm$ 17.01
Ennomos	11.28 $\pm$ 0.14	5.60 $\pm$ 1.52	4.06 $\pm$ 0.20	0.009 $\pm$ 0.010	20.24 $\pm$ 7.38
Iphinous	10.09 $\pm$ 0.37	6.54 $\pm$ 0.47	4.17 $\pm$ 0.12	0.020 $\pm$ 0.001	26.96 $\pm$ 3.48

Note. Here crP₁₀ is the 10 μm plateau peak position in continuum-removed space.

and Iphinous have peaks at $\sim 10.45 \mu\text{m}$, slightly short of the primary fayalite resonance band. If this feature is due to fayalite, the $\sim 0.05 \mu\text{m}$ difference in position could be due to the Trojan spectra being not fully in the volume scattering regime (see Figures 4 and 6 in Martin et al. 2022). Aneas, Diomedes, Ennomos, and Iphinous have peaks that roughly line up with the $11.4 \mu\text{m}$ fayalite peak, but the relative height of that feature to others is inconsistent with fayalite. Though noisy, Eurybates and Deipylos have small peaks near $10.8 \mu\text{m}$, very close to the unique $10.9 \mu\text{m}$ fayalite peak.

5.2.2. Pyroxene

There are two common pyroxenes we will discuss: enstatite and diopside. Enstatite is an orthopyroxene (Ca-poor), diopside is a clinopyroxene (Ca-rich), and both are Mg-rich end-members. Like olivine bands, pyroxene bands shift due to changes in composition, but the shift is not linear (e.g., Figure 2 in Chihara et al. 2002). As more work has been done to constrain the location of orthopyroxene resonance bands, we include the range of positions for enstatite with Mg# from 100 to 50 in our analysis (abbreviated En₁₀₀ and En₅₀, respectively). The primary En₁₀₀ bands are at 9.2 and $10.7 \mu\text{m}$, while the En₅₀ bands are at 9.5 and $10.5 \mu\text{m}$. Diopside has three primary resonances: 9.3 , 10.3 , and $11.4 \mu\text{m}$ (Koike et al. 2000). Hedenbergite is the Fe-rich clinopyroxene, and it has resonance features in roughly similar positions as diopside, but few hedenbergite emission spectra have been recorded in the literature. The reflectance spectra of hedenbergite samples prepared to mimic highly porous and fine-particulate regoliths have peaks at ~ 9.3 , 10.3 , and $11.6 \mu\text{m}$ (Martin et al. 2023).

As discussed in the previous section, many of the peaks possibly identified with fayalite are better fit by enstatite. Because enstatite has two strong bands, the presence of the $\sim 9.3 \mu\text{m}$ band paired with the $\sim 10.5 \mu\text{m}$ band leads to our identification of enstatite. All of the Trojans, aside from Deiphobus and Teucer, have peaks near both enstatite resonance bands. Using the shorter-wavelength primary enstatite band, we can estimate the Mg# associated with each peak. Achilles has a $9.21 \mu\text{m}$ peak, while Nestor and Patroclus have $9.15 \mu\text{m}$ peaks, which all indicate a Mg# of 90–100; Hektor, Agamemnon, Thestor, and Iphinous have peaks close to $9.3 \mu\text{m}$, indicating a Mg# of 80–90; and Aneas, Diomedes, Eurybates, and Deipylos have features at $\sim 9.5 \mu\text{m}$, indicating a

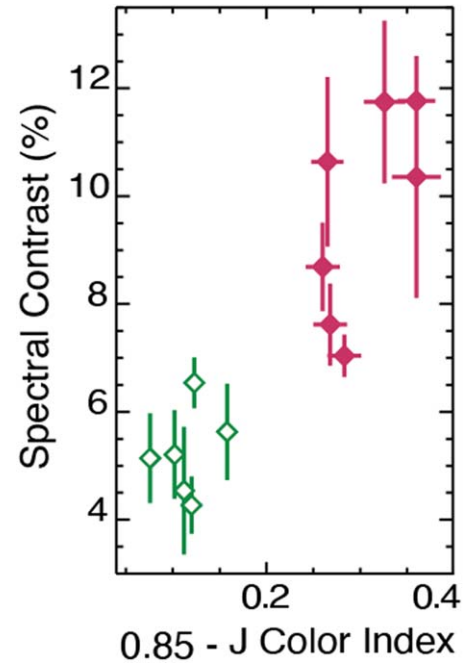


Figure 4. Spectral contrast of the 10 μm plateau plotted against the color index from $0.85 \mu\text{m}$ to the J band of the 11 Trojans presented here. The J band is measured from 1.1 to $1.4 \mu\text{m}$, centered on $1.25 \mu\text{m}$. Green symbols are less-red Trojans, and red symbols are red Trojans. The color index values are from Emery et al. (2011).

Mg# of 50–80. The longer-wavelength enstatite peak is in the correct location for the assigned Mg# in spectra of Patroclus, Nestor, Diomedes, Eurybates, Thestor, Deipylos, and Iphinous (see Chihara et al. 2002 for complete list of feature positions). If the $9.21 \mu\text{m}$ peak in Achilles' spectrum is due to enstatite, the second peak should occur near $\sim 10.6 \mu\text{m}$. However, we find Achilles' second peak at $\sim 10.45 \mu\text{m}$, which is not consistent with enstatite. We therefore lean toward an interpretation of the Achilles $10.45 \mu\text{m}$ peak as due to fayalite. Finally, Ennomos' spectrum in the $9 \mu\text{m}$ region is very noisy and may or may not have a peak around $9.2 \mu\text{m}$.

Many of the Trojan spectra exhibit features potentially associated with diopside but that can be accounted for by other

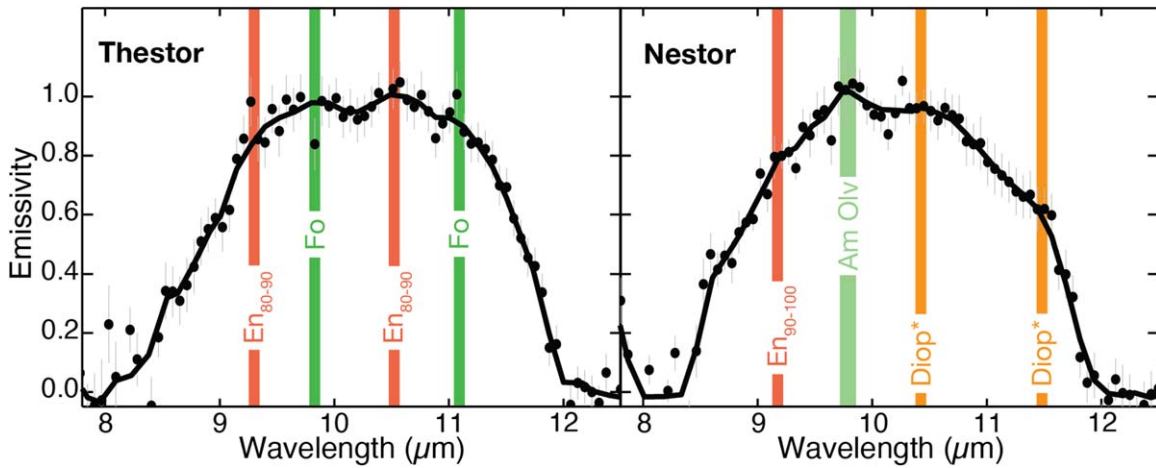


Figure 5. The 10 μm region spectra of Thestor and Nestor in continuum-removed space, with bars indicating the peak location of enstatite (En; red), forsterite (Fo; green), amorphous olivine (Am Oliv), and potential diopside (Diop; orange) features. The continuum is modeled as a line from D_8 to D_{12} and depicted as a gray dashed line (see Section 4.2), and the 10 μm plateau is scaled to span zero to 1 in emissivity space. We rebinned the spectra with a four bin width Gaussian smoothing function to highlight feature locations. An asterisk indicates a nonunique correlation to diopside.

Table 5
The Position of Peaks in the 10 μm Region Discussed in Section 5

Object	Position (μm)
Achilles	9.21, 9.77, 10.45, 11.10 ^a
Patroclus	9.15, 10.02, ^b 10.57, 11.10
Hektor	9.30, 10.0, 10.40, 11.10
Nestor	9.15, ^b 9.83, 10.26, 10.64, 11.55
Aneas	9.51, 9.95, 10.45, 11.10, 11.40
Agamemnon	9.30, 9.98, 10.45, 11.10
Diomedes	9.52, 9.90, 10.57, 11.13, 11.45 ^a
Deiphobus	11.10 ^a
Teucer	9.77, 11.10 ^a
Eurybates	8.90, ^b 9.40, ^b 10.10, 10.45, 10.80, ^b 11.50 ^a
Thestor	9.27, 9.80, ^b 10.45, 11.10
Deipylus	9.46, 10.08, 10.45, 10.80, ^b 11.10 ^b
Ennomos	9.77, 10.70, 11.10, ^b 11.26
Iphinous	9.28, ^a 9.89, 10.45, 11.10, ^b 11.26

Notes.

^a Indicates a shoulder.

^b Indicates a weak band.

minerals. The diopside resonance bands (at ~ 9.3 , 10.3 , and $11.4 \mu\text{m}$) have roughly equal spectral contrasts, so to make an identification, all three should be identified. Eurybates' spectrum is noisy but may have peaks near 9.4 , 10.1 , and $11.5 \mu\text{m}$. However, we interpret the $9.4 \mu\text{m}$ peak as a primary enstatite resonance, while the small 10.1 and $11.5 \mu\text{m}$ peaks can also be attributed to smaller enstatite resonance bands. Nestor is the only Trojan with a feature near $10.3 \mu\text{m}$, the location of a diopside resonance band. There does not appear to be a $9.3 \mu\text{m}$ equivalent in Nestor's spectrum, but the $11.55 \mu\text{m}$ peak could be accounted for with a solid solution between diopside and hedenbergite (see Figure 5).

5.2.3. Phyllosilicates

We found no unequivocal evidence that Trojans have phyllosilicates on their surfaces. Smectite has a broad resonance band around $9.9 \mu\text{m}$ (e.g., Koike & Shibai 1990). The Diomedes and Iphinous $9.9 \mu\text{m}$ peaks we attributed to enstatite in the previous section may be due to smectite.

However, Iphinous has a peak at $11.3 \mu\text{m}$, the region of a smaller enstatite band, so it likely has no serpentine on its surface. At longer wavelengths, phyllosilicates, including smectite, have resonance bands around 18 and $23 \mu\text{m}$ (Koike & Shibai 1990). Only Teucer has peaks near 18 and $23 \mu\text{m}$. As Teucer does not have peaks in the $10 \mu\text{m}$ region that would indicate phyllosilicates, the longer-wavelength peaks are likely due to a combination of anhydrous silicate bands (e.g., weak forsterite bands at 18.2 , 19.7 , 20.8 , 21.4 , and $23.7 \mu\text{m}$; Koike et al. 2003). Of note, these comparison phyllosilicates are terrestrial in origin. As phyllosilicates are complex and come in a wide range of compositions, the compositions of terrestrial phyllosilicates are not necessarily the same as those found in meteorites, for example.

The CM and CI meteorites are rich in nonterrestrial phyllosilicates. Laboratory-based studies outlined in Vernazza et al. (2012, 2013) and Izawa et al. (2021) looked at KBr-diluted meteorite spectra in the MIR and compared them to asteroid spectra, including the Trojans Hektor and Patroclus. Direct comparisons showed that the meteorite phyllosilicates do not match the Trojan emissivity spectra. The KBr acts as a proxy for regolith porosity, as it is transparent in the MIR. The spectra of KBr-diluted meteorites such as Ivuna (CI), Orgueil (CI), and Murray (CM2) show sharp triangular $10 \mu\text{m}$ features, whereas olivine- and pyroxene-rich meteorites tend to have more trapezoidal $10 \mu\text{m}$ plateaus with individual peaks that match those in some Trojan spectra, as described above. Though not all Trojan spectra have trapezoidal $10 \mu\text{m}$ plateaus, none have triangular $10 \mu\text{m}$ features like the phyllosilicates. As such, we see no indication that Trojans have phyllosilicates on their surfaces.

5.2.4. Amorphous Silicates

Amorphous olivine and pyroxene are glassy solids with the same ratios of Si, O, and Mg/Fe as in the crystalline olivine and pyroxene lattice, respectively. The spectrum of amorphous olivine has a broad peak around $9.8 \mu\text{m}$, while amorphous pyroxene has a broad peak in the $9.3 \mu\text{m}$ region (e.g., Figure 6). Achilles, Nestor, Teucer, Thestor, and Ennomos have broad features that peak near $9.8 \mu\text{m}$, indicative of amorphous olivine. Hektor, Agamemnon, Iphinous, and Thestor have a

Table 6
Summary of the Strength of Evidence for the Presence of Common Minerals and the Trojan Spectra from Individual Feature Analysis

Object	Olivine			Pyroxene		
	Forsterite	Fayalite	Am. ^a	Enstatite	Diopside	Am. ^a
Achilles	Strong	Strong	Strong	Strong	NE	NE
Patroclus	Strong	Weak	NE	Strong	NE	NE
Hektor	Strong	Weak	NE	Strong	NE	Weak
Nestor	NE	Weak	Strong	Strong	Weak	NE
Agamemnon	Strong	Weak	NE	Strong	NE	Weak
Aneas	Strong	Weak	NE	Strong	NE	NE
Diomedes	NE	Weak	NE	Strong	NE	NE
Deiphobus	Strong	NE	NE	NE	NE	NE
Teucer	Strong	NE	Strong	NE	NE	NE
Eurybates	Strong	Weak	NE	Strong	NE	NE
Thestor	Strong	Weak	Strong	Strong	NE	Weak
Deipylos	Strong	Weak	NE	Strong	NE	NE
Ennomos	Strong	Weak	NE	Strong	NE	NE
Iphinous	Strong	Weak	Strong	Strong	NE	Weak

Notes. Common minerals include crystalline and amorphous (am.) olivine and pyroxene. Strong evidence for a mineral is denoted with “strong,” weak or potential evidence is denoted with “weak,” and no evidence of a mineral is denoted with “NE.” Evidence of amorphous minerals outside of the peak analysis is discussed in Section 5.2.4. Note that strong evidence does not necessarily equate to a strong feature(s).

^a This table only discusses evidence from specific peak analysis. There is strong evidence for amorphous material in all objects from spectral modeling; see Section 5.2.4.

broad $9.3\ \mu\text{m}$ feature that may be due to amorphous pyroxene. The position of this feature is also consistent with crystalline enstatite, but we cannot rule out a mixture of both crystalline and amorphous enstatite on their surfaces.

For a qualitative estimate of the crystalline fraction of silicates (f_{cryst} ; the fraction of silicates that are crystalline versus amorphous in a mixture) on Trojan surfaces, we linearly mixed the spectra of amorphous and crystalline silicates and multiplied the spectral mixture by 0.1 to reduce the spectral contrast to more closely match that of the Trojans. The $10\ \mu\text{m}$ plateau is very rounded in each Trojan spectrum, aside from Thestor and Hektor being more trapezoidal. In order to achieve the average roundness of a Trojan spectrum with amorphous and crystalline silicate spectra, the f_{cryst} value needs to be at most 20% (Figure 6). As the amorphous component increases, sharp crystalline features decrease in spectral contrast for both olivine and pyroxene (Figure 6(a)). When f_{cryst} is $\gtrsim 20\%$, the sharp crystalline features have spectral contrasts that are too high to be consistent with most Trojan spectra.

Additionally, we looked at the mixture of amorphous olivine and pyroxene (Figure 6(b)). When there is more amorphous olivine, the whole $10\ \mu\text{m}$ plateau is shifted toward longer wavelengths, which looks more like the Patroclus spectrum. We acknowledge that the simple mixture of crystalline pyroxene and olivine with amorphous olivine is not enough to accurately reproduce the Patroclus spectrum. There must be additional material to suppress the shorter-wavelength silicate peaks. We have not found any combination of amorphous and crystalline silicate spectra that can reproduce the skew we see in the $10\ \mu\text{m}$ plateau (taller peaks at longer wavelengths). Incorporating fayalite helps to pull the $10\ \mu\text{m}$ plateau a bit toward longer wavelengths (Figure 6(c)) but not enough to match up with the Trojan spectra with skews higher than $\sim 14\%$ (Table 4).

From our simple mixture analysis, Hektor’s and Thestor’s f_{cryst} values are $\sim 20\%$ (i.e., 20% of the silicates on their surfaces are crystalline). Ennomos and Deipylos also have potentially trapezoidal $10\ \mu\text{m}$ plateaus and may have f_{cryst}

values of 20% as well. Deiphobus and Teucer have the most rounded $10\ \mu\text{m}$ plateaus, so their f_{cryst} values are likely below $\sim 10\%$. Finally, the estimated f_{cryst} for the remaining Trojans is closer to 10%.

5.3. Comets

Spectroscopically, comets are suitable analogs for Trojans due to their strong emissivity features, red-sloped VNIR spectra, and low thermal inertia. Like Thestor, Deiphobus, Teucer, Achilles, Diomedes, Hektor, Aneas, and Agamemnon, comet nuclei are typically classified as D type (e.g., Campins et al. 2006; DeMeo & Binzel 2008). Though Trojans do not show spectral evidence of organics or ices on their surfaces, like that found in comet comae (e.g., Farnham et al. 2007; Feaga et al. 2007; Harker et al. 2007), and Trojans have no known comae (Perna et al. 2018), the spectra of coma silicates are still useful for comparison, as they are sourced from the nuclei. Typically, coma silicates are dominated by amorphous phases (see Section 6.2; Wooden 2002; Kelley & Wooden 2009), though crystalline silicates have been detected (e.g., Harker et al. 2005, 2007; Sugita et al. 2005; Lisse et al. 2006).

In the MIR, spectral similarities between the emissivity spectra of comet comae like C/1995 O1 (Hale–Bopp) and 29P/Schwassmann–Wachmann 1 (Emery et al. 2006) and comet nuclei like 10P/Tempel 2 (Kelley et al. 2017) and Hektor, Aneas, and Agamemnon have been previously noted. The best cometary analog for Hektor is the nucleus of Jupiter-family comets (JFC), Tempel 2 (Kelley et al. 2017), which has a fine-particulate regolith of Mg-rich and Mg/Fe mixed silicates. As JFCs are thought to originate from the primordial Kuiper Belt, this spectral connection strengthens the hypothesis that Trojans formed in the primordial Kuiper Belt as well.

As with the previous Trojan spectra, we find many similarities between comet spectra and the Trojan spectra presented here. Spectra of both comae and nuclei have pronounced $10\ \mu\text{m}$ plateaus, a dip in the $\sim 13\text{--}15\ \mu\text{m}$ region, and broad peaks near 18 and $24\ \mu\text{m}$ (Figure 7). The $10\ \mu\text{m}$

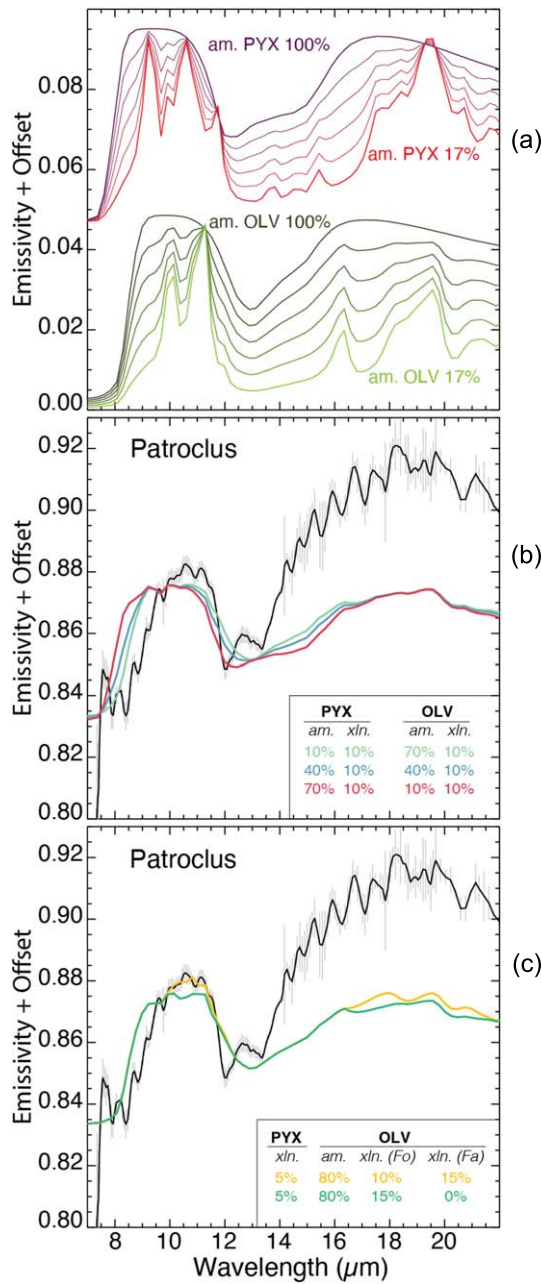


Figure 6. (a) Mixtures of spectra of amorphous and crystalline minerals: pyroxene (PYX) on top in reds, and olivine (OLV) on the bottom in greens. Each suite has the spectra of the amorphous (am.) component mixed with the 0%–83% crystalline (xln.) component. Fully amorphous spectra are in darker shades. (b) Patroclus spectrum with three spectral mixtures overlaid: 10% xln. PYX and OLV, 10% am. PYX, and 70% am. OLV in green; 10% xln. PYX and OLV and 40% am. PYX and OLV in blue; and 10% xln. PYX and OLV, 70% am. PYX, and 10% am. OLV in red. Note the position shift from am. OLV–dominant spectra to am. PYX–dominant spectra. (c) Patroclus spectrum with two spectral mixtures overlaid: 5% xln. PYX, 10% xln. forsterite (Fo), 15% xln. fayalite (Fa), and 80% am. OLV in yellow and 5% xln. PYX, 15% xln. Fo, 0% xln. Fa, and 80% am. OLV in green. Note the difference in the general shape of the 11 μm region when fayalite is incorporated.

plateau spectral contrast is much smaller for comet nuclei than for comae, making the spectra of comet nuclei more suitable analogs for Trojans. Additionally, like Trojan spectra, many comets show spectral evidence of amorphous silicates that produce broad rounded emissivity peaks in the 10 μm region (e.g., Harker et al. 2007, 2011; Kelley & Wooden 2009).

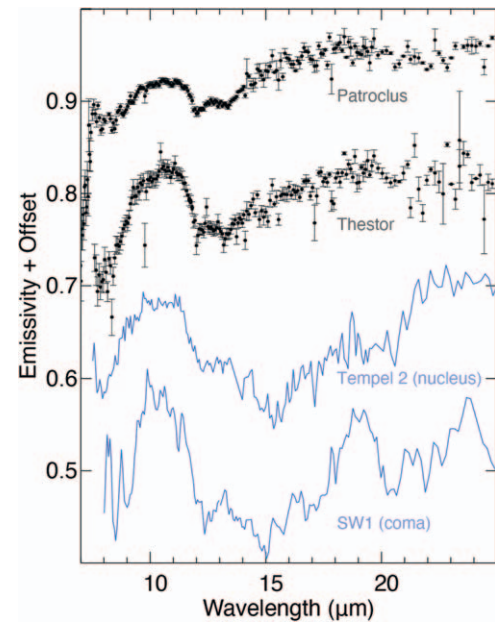


Figure 7. Spectra of Patroclus and Thetor plotted on top of spectra of comet nucleus 10P/Tempel 2 and comet coma 29P/Schwassmann–Wachmann 1. The comet spectra are from Kelley et al. (2017).

Although the spectral shapes are consistent, the Trojan 10 μm plateau spectral contrast is much smaller compared to comet comae and nuclei. A possible explanation is the prevalence of opaques on the Trojans, like amorphous carbon, which dampen silicate emission features in the MIR (e.g., Donaldson Hanna et al. 2019; Beck et al. 2022). A combination of amorphous silicates and amorphous carbon were found to produce good matches to VNIR Trojan spectra (Cruikshank et al. 2001; Emery & Brown 2004; Emery et al. 2011).

6. Discussion

Thus far, we have shown that the MIR spectra of Trojan asteroids are like the spectra of comet comae and nuclei, identified common silicate features in their spectra, and estimated the crystalline fraction and regolith porosity. The analysis and comparisons (see Section 5) indicate that Trojan surfaces are covered in a highly porous regolith of fine-particulate, dominantly amorphous, and Mg-rich silicates mixed with opaque material. Now we circle back to the question of where in the solar system the Trojans formed and how their surface composition adds to their formation story. As a reminder, there are two Trojan formation hypotheses: in the primordial Kuiper Belt and near their current position. Many aspects of Trojan dynamics and population statistics are consistent with dynamical models like the Nice model, favoring a Kuiper Belt origin. Such dynamical models explain not only how the Trojans might have traveled inward but also the distribution of giant planets and KBOs.

6.1. Regolith Porosity

First, we note that the depth of porosity measurable via MIR spectral techniques is about a millimeter. Regolith porosity may decrease with increasing depth due to compaction of the overlying material, which is also an important factor for volatile retention (e.g., Schorghofer 2016). As such, our regolith porosity estimations are distinct from bulk density

calculations. Three Trojans have measured bulk densities: Hektor ($\sim 1 \text{ g cm}^{-3}$; Marchis et al. 2014), Patroclus ($\sim 0.8 \text{ g cm}^{-3}$; Marchis et al. 2006; Mueller et al. 2010; Buie et al. 2015; Berthier et al. 2020), and Eurybates ($1.1 \pm 0.3 \text{ g cm}^{-3}$; Brown et al. 2021). These bulk density values are suggestive of a high degree of volatiles, such as water ice (e.g., Mueller et al. 2010). An additional five Trojans will have measured bulk densities after NASA’s Lucy mission flybys from 2027 to 2033 (Levison et al. 2021; Olkin et al. 2021).

Identifying the Trojan MIR spectral scattering regime is an important step for choosing the appropriate laboratory analog, which leads to more accurate mineral identification and stronger connections to other small bodies in the solar system. The regolith porosity we estimate on Trojan surfaces ($\sim 80\%$ for Eurybates and $\sim 90\%$ for the rest) from MIR spectral analysis is the probable cause for their seemingly anomalous MIR spectra. There are several scenarios that may have caused such high regolith porosities.

One scenario draws from the closest spectral analog to Trojans: comet nuclei. On comet nuclei, regolith porosities are $\sim 33\%$ – 76% (e.g., Kossacki 2021). Such regolith porosities result from lofted dust particles entrained in outgassing events that fell back onto comet surfaces. It is possible that Trojans have slowly sublimated volatiles from their surfaces and near-surfaces, resulting in lofted dust particles that eventually fell back to the surface. Over time, the lofting and falling of dust could have created the extreme “fairy-castle” structures hypothesized by Emery et al. (2006). Additionally, the bulk regolith porosities of Hektor, Patroclus, and Eurybates are indicative of subsurface volatiles (e.g., Mueller et al. 2010), which would be the source of sublimation and outgassing events. As Trojan regoliths have $\sim 90\%$ regolith porosities, it is possible that Trojan surfaces can accumulate more dust compared to comet nuclei because comets are resurfaced with each orbit. Due to the spectral similarities between Trojans and comets and a simple reasonable mechanism, this scenario is favored as a primary cause for high regolith porosity.

A similar scenario that can describe high regolith porosities, proposed by Yang et al. (2013), describes Trojan surfaces covered in a layer of salty lag deposits left by frozen brine that had been exposed by periodic impacts. Fine-particulate silicates could remain on Trojan surfaces embedded in a salty matrix. And because most salts are highly transparent in the MIR, the spectral signature would only show the embedded silicates. The Yang et al. (2013) hypothesis requires that Trojans have hydrated silicates below their regoliths that are periodically exposed by recent impacts (i.e., after the Late Heavy Bombardment). They explained that hydrated silicate spectral features may be shielded by thick evaporate layers. A mixture of opaque material and olivine in a salt matrix (over 90 wt% salt) can account for NIR spectral slopes and the MIR $10 \mu\text{m}$ plateau in Trojan spectra. However, the albedo of a salty regolith is much higher than the Trojans (e.g., Izawa et al. 2021).

An additional process, impact gardening, results in high regolith porosities, as is the case for high porosity ($\gtrsim 80\%$) in the lunar regolith (e.g., Arnold 1975; Hapke & Sato 2016; Pieters & Noble 2016). Interestingly, the MIR spectra of the Moon and Trojans are significantly different from each other (e.g., Shirley & Glotch 2019), which suggests a surface scattering-dominated spectral regime on the Moon. One explanation for the different MIR spectra is that the lunar

regolith may be made of larger particles compared to Trojan regoliths. Further, on low-gravity objects like Trojans, impact gardening could result in higher degrees of regolith porosity, as the gravitational pull is significantly less than on the Moon, which also renders less compaction. As impact gardening is important throughout the solar system, it could be a major contributor to the fluffy top layer of Trojan regoliths.

Recent work suggests that a mixture of hyperfine opaque material and olivine results in spectral similarities to P- and D-type asteroids like Trojans (Sultana et al. 2023). The spectra of opaques (FeS and anthracite) and olivine mixtures show prominent $10 \mu\text{m}$ plateaus, masked NIR silicate features, and low albedos. The presence of hyperfine opaque material on Trojan surfaces would affect the total regolith porosity estimation but not the mineralogy interpretations.

6.2. Mineralogy and Composition

The lack of phyllosilicate evidence in Trojan spectra is consistent with a Kuiper Belt origin. In addition to no phyllosilicate features being identified in MIR spectra, no phyllosilicate features have been detected in the VNIR spectra (at 0.7 and $3 \mu\text{m}$) of Trojans (e.g., Cruikshank et al. 2001; Emery & Brown 2003). Under theories like the Grand Tack model, many asteroids that formed within ~ 13 au should have accreted with abundant volatiles (Walsh et al. 2011). The lack of phyllosilicate evidence in Trojan spectra is consistent with a Kuiper Belt origin, though it does not entirely rule out an inner solar system origin.

The composition of olivine and pyroxene can be indicative of the solar system formation region, with Mg-rich forsterite and enstatite expected to form in the inner solar system, although radial mixing of silicates likely occurred in the early solar system (e.g., Henning 2010, and references therein). Dust from the comet 81P/Wild 2 coma was collected by NASA’s Stardust Mission (Brownlee 2014) and was dominated by silicates with Fo/En_{90–75} and Fo/En_{75–50} compositions, as well as minor amounts of Fo/En_{100–90} (Frank et al. 2014). These three Mg# ranges are distinguished in cometary studies because the crystallization environment for each range is distinctly different. An important commonality among cometary material (i.e., primitive silicates) is a wide range of Mg#s that have been found in individual objects. Though the MIR spectra of comet comae have not yielded evidence of silicates with Mg#s below ~ 90 , astronomers predict that they are there but spectrally shielded (Wooden et al. 2017). Our results indicate that crystalline olivine in Trojan regoliths tends to be forsteritic (i.e., Mg-rich), while crystalline pyroxene is Mg-rich on the surfaces of Achilles, Patroclus, Nestor, Agamemnon, Thestor, and Iphinous and intermediate on the surfaces of Aeneas, Diomedes, Eurybates, and Deipylos. The forsteritic olivine is consistent with a variety of origin theories, while the range of pyroxene compositions is only consistent with cometary, i.e., Kuiper Belt, origins.

Amorphous silicates are common materials in the MIR spectra of cometary comae (e.g., Harker & Desch 2002; Harker et al. 2007, 2011; Woodward et al. 2011) and the Trojan spectra presented here. The MIR spectral investigations of comets produce f_{cryst} values from 79% (Hale-Bopp at perihelion; Harker et al. 2004) to $\sim 14\%$ (C/2005 N3 (Lulin); Woodward et al. 2011) but typically fall in the 20%–70% range (Wooden et al. 2017). Ultracarbonaceous Antarctic micro-meteorites (UCAMMs) are thought to be cometary material

(Wooden et al. 2017, and references therein). The UCAMMs are dominated by carbon and contain abundant amorphous material called glass with embedded metal and sulfides, as well as crystalline silicates with a range of Mg#s, and they have an f_{cryst} value of $\sim 25\%$ (e.g., Dobrică et al. 2012). Models of crystalline and amorphous silicate condensation in the protoplanetary disk predict that in the main-belt region (2–5 au), crystalline silicate phases make up 75%–90% of the silicate material, whereas silicates that condensed at distances greater than ~ 10 au should have f_{cryst} values of $\sim 0\%$ –30% (Gail 2004; Henning 2010). Some observed discrepancies between model predictions and measurements suggest a high degree of radial mixing in the protoplanetary disk and/or sources from outside the protoplanetary disk (e.g., Wooden et al. 2017). While the origin of amorphous silicates detected in comet spectra is an ongoing debate, a low f_{cryst} value ($\lesssim 30\%$) is indicative of cometary material, even though high f_{cryst} values ($\gtrsim 75\%$) are also observed. Nevertheless, the f_{cryst} values we determined for Trojans of $\sim 20\%$ and lower are consistent with values from the spectra of most comet comae, UCAMMs, and condensation models for accretion beyond 10 au. As such, Trojan f_{cryst} values indicate an outer solar system origin, consistent with the primordial Kuiper Belt region.

6.3. A Special Consideration for Eurybates

Eurybates is a particularly interesting asteroid and the first Trojan target of the Lucy mission. Eurybates is the parent body of the largest confirmed asteroid family in the Trojan population (Brož & Rozehnal 2011; Nesvorný 2015). In the VNIR, Eurybates has a less-red slope and is classified as a C-type asteroid (DeMeo et al. 2009). Though C types are not rare among MBAs, only $\sim 10\%$ of the Trojan population consists of C types, and most are in the Eurybates family (e.g., Fornasier et al. 2007; DeMeo et al. 2009). Additionally, Eurybates appears to have a spectrally heterogeneous surface, leading researchers to hypothesize an irregular distribution of particle sizes and regolith compaction across the surface (Souza-Feliciano et al. 2020).

Eurybates is the only object with a potential CF, and it may have the least porous regolith. The compositional similarity of Eurybates' surface compared to other Trojans in this study (see Table 6) leads us to postulate that spectral differences arise due to surface texture variability (i.e., particle size and regolith porosity). As Eurybates is the largest member of an asteroid family, it has likely undergone catastrophic resurfacing more recently than other Trojans. Not only would this event expose younger material, it may have also knocked off much of its original fine-particulate regolith. In Section 6.1., we suggest that regolith porosity on Trojan surfaces is largely due to slowly sublimating volatiles over long time periods. If true, the event that created the Eurybates family may have occurred after many of the volatiles had already been depleted from the surface. This leaves impact gardening, landslides, and possibly fine-particulate debris related to the creation of the Eurybates family as the remaining sources for regolith porosity. As such, we expect regions of high porosity to be concentrated near the base of steep slopes where landslides would occur and on redder surfaces that would have been less affected by the breakup. A spotty regolith porosity across Eurybates' surface is consistent with observations in the NIR (e.g., Souza-Feliciano et al. 2020) and will be verifiable by images and spectra returned from the Lucy mission.

6.4. The Trojan Color Bimodality Conundrum

As mentioned in Section 4.2, there is a correlation between the spectral contrast of the $10\ \mu\text{m}$ feature and VNIR color. This correlation was not expected because there are two distinct color categories in the VNIR hypothesized to be caused by compositional differences (e.g., Wong & Brown 2016), and the $10\ \mu\text{m}$ feature spectral contrasts are not correlated with silicate mineralogy as detected in the MIR (see Table 6). Furthermore, in the VNIR, color groups are bimodal, whereas the $10\ \mu\text{m}$ feature spectral contrasts show a continuous trend (Figure 4), which begs the question: why are the spectral contrasts of the $10\ \mu\text{m}$ feature correlated to the VNIR color bimodality? Here we posit three potential explanations.

1. Red Trojans may be more susceptible to volatile loss compared to less-red Trojans. Martin et al. (2022, 2023) showed that the spectral contrast of the $10\ \mu\text{m}$ feature is correlated to regolith porosity: the higher the regolith porosity, the larger the spectral contrast. We hypothesize that the primary cause of regolith porosity on Trojans is a combination of sublimating volatiles and impact gardening. Since impact gardening is ubiquitous, the effect should be as well. So, objects with redder slopes may somehow be more susceptible to volatile sublimation compared to objects with less-red slopes. A slight preference for red Trojans to sublimate more readily may lead to the continuous trend in $10\ \mu\text{m}$ feature spectral contrasts.
2. Alternatively (or compounding), less-red Trojans may have a greater ability to create thick sublimation lags, leading to less sublimation compared to red Trojans. This would also account for the continuous trend in $10\ \mu\text{m}$ feature spectral contrasts that is correlated with a bimodal color distribution.
3. Less-red surfaces may be fresher. The Wong & Brown (2016) hypothesis leads to the prediction that all Trojans are internally less red and collisional fragments in the Trojan population should be less red. If true, collisional fragments like those in the Eurybates family should have “fresher” surfaces with lower regolith porosities compared to both red and older less-red Trojans. As such, the smaller/dynamically young less-red Trojans should generally have the lowest $10\ \mu\text{m}$ feature spectral contrast values and show evidence of a CF, as potentially seen in Eurybates' spectrum.

7. Conclusions and Future Work

We present MIR spectra of 14 Trojan asteroids measured by the SST. Modeled sizes, albedos, and temperatures are consistent with prior estimations of each Trojan. Through parameterization of spectral features and comparison to laboratory and remote-sensing spectra, the results show that the Trojans have highly porous regoliths of fine-particulate crystalline (forsterite, enstatite, and possibly a small amount of fayalite and diopside) and amorphous silicates. Our compositional results (abundant forsterite and potential minor amounts of fayalite, a wide range of pyroxene Mg#s, evidence of amorphous silicates, and no evidence of phyllosilicates) are consistent with the Nice model, which predicts the Trojans forming in the primordial Kuiper Belt.

As we have shown, a detailed analysis of MIR spectra can yield robust compositional results. This is especially important for asteroids with featureless NIR spectra, like Trojans. New

MIR spectra of asteroids using the James Webb Space Telescope and paramount findings from the Lucy mission to the Trojans will undoubtedly lead to a deeper understanding of compositional distribution among primitive asteroids and the history of the solar system.

Acknowledgments

The authors would like to thank Sean Lindsay for invaluable discussions regarding the inception and methodology of this work. Thank you to the two anonymous reviewers for their constructive feedback. We would also like to acknowledge that this work was primarily done in northern Arizona, at the base of the San Francisco Peaks, on homelands sacred to Native Americans throughout the region. Additional work was done in eastern Tennessee, on the traditional land of the Tsalagi and Tsoyaha people. This work is based (in part) on observations made with the Spitzer Space Telescope, which was operated by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with NASA. This work is supported by the National Aeronautics and Space Administration's Science Mission Directorate Research and Analysis Solar System Workings program NH19ZDA001N.

ORCID iDs

Audrey C. Martin  <https://orcid.org/0000-0003-3402-1339>

Joshua P. Emery  <https://orcid.org/0000-0001-9265-9475>

References

- Arnold, J. R. 1975, *Moon*, **13**, 159
- Beck, P., Sultana, R., Poch, O., Quirico, E., & Schmitt, B. 2022, *LPSC*, **53**, 1427
- Berthier, J., Descamps, P., Vachier, F., Normand, J., & Maquet, L. 2020, *Icar*, **352**, 113990
- Brown, M. E., Levison, H. F., Noll, K. S., et al. 2021, *PSJ*, **2**, 170
- Brownlee, D. 2014, *AREPS*, **42**, 179
- Brož, M., & Rozehnal, J. 2011, *MNRAS*, **414**, 565
- Buie, M. W., Olkin, C. B., Merline, W. J., et al. 2015, *AJ*, **149**, 113
- Campins, H., Ziffer, J., Licandro, J., et al. 2006, *AJ*, **132**, 1346
- Chihara, H., Koike, C., Tsuchiyama, A., Tachibana, S., & Sakamoto, D. 2002, *A&A*, **391**, 267
- Christensen, P. R., Bandfield, J. L., Hamilton, V. E., et al. 2000, *JGR*, **105**, 9735
- Clark, B. E., Ockert-Bell, M. E., Cloutis, E. A., et al. 2009, *Icar*, **202**, 119
- Cruikshank, D. P., Dalle Ore, C., Roush, T. L., et al. 2001, *Icar*, **153**, 348
- DeMeo, F., & Binzel, R. P. 2008, *Icar*, **194**, 436
- DeMeo, F. E., & Carry, B. 2013, *Icar*, **226**, 723
- DeMeo, F. E., Binzel, R. P., Silvan, S. M., & Bus, S. J. 2009, *Icar*, **202**, 160
- Dobrica, E., Engrand, C., Leroux, H., et al. 2012, *GeCoA*, **76**, 68
- Donaldson Hanna, K. L., Greenhagen, B. T., Patterson, W. R., III, et al. 2017, *Icar*, **283**, 326
- Donaldson Hanna, K. L., Schrader, D. L., Cloutis, E. A., et al. 2019, *Icar*, **319**, 701
- Emery, J. P., & Brown, R. H. 2003, *Icar*, **164**, 104
- Emery, J. P., & Brown, R. H. 2004, *Icar*, **170**, 131
- Emery, J. P., Cruikshank, D. P., & Van Cleve, J. 2006, *Icar*, **182**, 496
- Emery, J. P., Burr, D. M., & Cruikshank, D. P. 2011, *AJ*, **141**, 25
- Farnham, T. L., Wellnitz, D. D., Hampton, D. L., et al. 2007, *Icar*, **191**, 146
- Feaga, L. M., A'Hearn, M. F., Sunshine, J. M., Groussin, O., & Farnham, T. L. 2007, *Icar*, **190**, 345
- Fernández, Y. R., Emery, J. P., Cruikshank, D. P., & Stansberry, J. A. 2003, *AJ*, **126**, 1563
- Fleming, H. J., & Hamilton, D. P. 2000, *Icar*, **148**, 479
- Fornasier, S., Dotto, E., Marzari, F., et al. 2007, *Icar*, **190**, 622
- Frank, D. R., Zolensky, M. E., & Le, L. 2014, *GeCoA*, **142**, 240
- Gaffey, M. J., Bell, J. F., Brown, R. H., et al. 1993, *Metic*, **28**, 161
- Gail, H. P. 2004, *A&A*, **413**, 571
- Gomes, R., Levison, H. F., Tsiganis, K., & Morbidelli, A. 2005, *Natur*, **435**, 466
- Gradie, J., & Veverka, J. 1980, *Natur*, **283**, 840
- Grav, T., Mainzer, A. K., Bauer, J., et al. 2011, *ApJ*, **742**, 40
- Grimm, R. E., & Mcsween, H. Y., Jr. 1989, *Icar*, **82**, 244
- Hamilton, V. E. 2010, *ChEG*, **70**, 7
- Hapke, B., & Sato, H. 2016, *Icar*, **273**, 75
- Harker, D. E., & Desch, S. J. 2002, *ApJ*, **565**, 109
- Harker, D. E., Wooden, D. H., Woodward, C. E., & Lisse, C. M. 2004, *ApJ*, **615**, 1081
- Harker, D. E., Woodward, C. E., & Wooden, D. H. 2005, *Sci*, **310**, 278
- Harker, D. E., Woodward, C. E., Wooden, D. H., Fisher, R. S., & Trujillo, C. A. 2007, *Icar*, **190**, 432
- Harker, D. E., Woodward, C. E., Kelley, M. S., et al. 2011, *AJ*, **141**, 26
- Harris, A. W. 1998, *Icar*, **131**, 291
- Henning, T. 2010, *ARA&A*, **48**, 21
- Hiroi, T., Zolensky, M. E., Pieters, C. M., et al. 1996, *M&PS*, **31**, 321
- Hofmeister, A. M., & Pitman, K. M. 2007, *PCM*, **34**, 319
- Houck, J. R., Roelling, T. L., van Cleve, J., et al. 2004, *ApJS*, **154**, 18
- Hunt, G. R., & Logan, L. M. 1972, *ApOpt*, **11**, 142
- Izawa, M. R. M., King, P. L., Vernazza, P., Berger, J. A., & McCutcheon, W. A. 2021, *Icar*, **359**, 114328
- Jewitt, D. C., & Luu, J. X. 1990, *AJ*, **100**, 933
- Kelley, M. S., & Wooden, D. H. 2009, *P&SS*, **57**, 1133
- Kelley, M. S. P., Woodward, C. E., Gehr, R. D., et al. 2017, *Icar*, **284**, 344
- Koike, C., & Shibai, H. 1990, *MNRAS*, **246**, 332
- Koike, C., Tsuchiyama, A., Shibai, H., et al. 2000, *A&A*, **363**, 1115
- Koike, C., Chihara, H., Tsuchiyama, A., et al. 2003, *A&A*, **399**, 1101
- Kossacki, K. J. 2021, *Icar*, **368**, 114613
- Lane, M. D., Glotch, T. D., Dyar, M. D., et al. 2011, *JGRE*, **116**, E08010
- Levison, H. F., Shoemaker, E. M., & Shoemaker, C. S. 1997, *Natur*, **385**, 42
- Levison, H. F., Morbidelli, A., Vokrouhlicky, D., & Bottke, W. F. 2008, *AJ*, **136**, 1079
- Levison, H. F., Marchi, S., Noll, K., Olkin, C., & Statler, T. S. 2021, in 2021 IEEE Aerospace Conf. Proc. (Piscataway, NJ: IEEE), **1**
- Licandro, J., Hargrove, K., Kelley, M., et al. 2012, *A&A*, **537**, A73
- Lim, L. F., Emery, J. P., & Moskovitz, N. A. 2011, *Icar*, **213**, 510
- Lisse, C. M., VanCleve, J., Adams, A. C., et al. 2006, *Sci*, **313**, 635
- Marchis, F., Durech, J., Castillo-Rogez, J., et al. 2014, *ApJL*, **783**, L37
- Marchis, F., Hestroffer, D., Descamps, P., et al. 2006, *Natur*, **439**, 565
- Martin, A. C. 2022, PhD thesis, Northern Arizona Univ.
- Martin, A. C., Emery, J. P., & Loeffler, M. J. 2022, *Icar*, **378**, 114921
- Martin, A. C., Emery, J. P., & Loeffler, M. J. 2023, *Icar*, **397**, 115507
- Marzari, F., & Scholl, H. 1998, *Icar*, **131**, 41
- Morbidelli, A., Levison, H. F., Tsiganis, K., & Gomes, R. 2005, *Natur*, **435**, 462
- Mueller, M., Marchis, F., Emery, J. P., et al. 2010, *Icar*, **205**, 505
- Mustard, J. F., & Hays, J. E. 1997, *Icar*, **125**, 145
- Nesvorný, D. 2015, *AJ*, **150**, 73
- Nesvorný, D., Vokrouhlický, D., & Morbidelli, A. 2013, *ApJ*, **768**, 45
- Olkin, C. B., Levison, H. F., Vincent, M., et al. 2021, *PSJ*, **2**, 172
- Perna, D., Bott, N., Hromakina, T., et al. 2018, *MNRAS*, **475**, 974
- Pieters, C. M., & Noble, S. K. 2016, *JGRE*, **121**, 1865
- Rivkin, A. S., Davies, J. K., Johnson, J. R., et al. 2003, *M&PS*, **38**, 1383
- Rivkin, A. S., Volquardsen, E. L., & Clark, B. E. 2006, *Icar*, **185**, 563
- Romanishin, W., & Tegler, S. C. 2018, *AJ*, **156**, 19
- Salisbury, J. W., & Wald, A. 1992, *Icar*, **96**, 121
- Schorghofer, N. 2016, *Icar*, **276**, 88
- Shirley, K. A., & Glotch, T. D. 2019, *JGRE*, **124**, 970
- Souza-Feliciano, A. C., De Pra, M., Pinilla-Alonso, N., et al. 2020, *Icar*, **338**, 113463
- Sugita, S., Ootsubo, T., Kadono, T., et al. 2005, *Sci*, **310**, 274
- Sultana, R., Poch, O., Beck, P., et al. 2023, *Icar*, **395**, 115492
- Takir, D., & Emery, J. P. 2012, *Icar*, **219**, 641
- Tsiganis, K., Gomes, R., Morbidelli, A., & Levison, H. F. 2005, *Natur*, **435**, 459
- Vernazza, P., Delbo, M., King, P. L., et al. 2012, *Icar*, **221**, 1162
- Vernazza, P., Fulvio, D., Brunetto, R., et al. 2013, *Icar*, **225**, 517
- Vernazza, P., Castillo-Rogez, J., Beck, P., et al. 2017, *AJ*, **153**, 72
- Vincent, R. K., & Hunt, G. R. 1968, *ApOpt*, **7**, 53
- Walsh, K. J., Morbidelli, A., Raymond, S. N., O'Brian, D. P., & Mandell, A. M. 2011, *Natur*, **475**, 206
- Werner, M. W., Roellig, T. L., Low, F. J., et al. 2004, *ApJS*, **154**, 1
- Wong, I., & Brown, M. E. 2016, *AJ*, **152**, 90
- Wooden, D. H. 2002, *EM&P*, **89**, 247
- Wooden, D. H., Ishii, H. A., & Zolensky, M. E. 2017, *RSPTA*, **375**, 20160260
- Woodward, C. E., Jones, T. J., Brown, B., et al. 2011, *AJ*, **141**, 181
- Yang, B., & Jewitt, D. 2011, *AJ*, **141**, 95
- Yang, B., Lucey, P., & Glotch, T. 2013, *Icar*, **223**, 359